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Abstract

Maize (*Zea mays* ssp. *mays*) originated in Mexico and Central America and belongs to the tribe Maydaceae of the family Poaceae. It possesses over 32,000 genes on ten chromosomes with a genome size of 2.3 gigabase. Among various theories, teosinte (*Z. mays* ssp. *parviglumis*) is the most likely progenitor of maize. Genetic loci such as *teosinte branched 1* (*tb1*) and *teosinte glume architecture 1* (*tga1*) have played pivotal role in transforming teosinte to modern maize. From the centre of origin, maize later diffused to different parts of the world including America, Europe, Africa and Asia. Genetic resources especially wild relatives and landraces harbour novel alleles/genes for imparting resistance/tolerance to several biotic/abiotic stresses and improvement of productivity and nutritional quality traits. Among wild relatives, teosintes and *Tripsacum* are native to Mexico and Central America, while *Coix*, *Chionachne*, *Sclerachne*, *Trilobachne* and *Polytoca* originated in Southeast Asia. Cross compatibility between maize and teosinte generates hybrids, although the extent of success depends on teosinte species. While generation of crosses between maize and *Tripsacum* species is difficult, Asiatic genera possess high degree of cross incompatibility with maize. Several factors or loci that regulate cross compatibility between different genera or species have been identified and play pivotal role in introgression of genes from wild relatives. Landrace accessions possessing unexplored alleles/genes serve as valuable donors for important traits. The chapter deals with different theories of maize origin and recent understanding on diffusion of maize worldwide, besides providing classifications of wild relatives and their cross compatibility with maize. The importance of wild relatives and maize

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landrace accessions in providing valuable genes for various important traits has been discussed.

Keywords

Characterization • Landrace • Teosinte • *Tripsacum* • Wild relatives • *Zea mays*

4.1 Introduction

Maize (*Zea mays* ssp. *mays*) belongs to the tribe Maydaceae, family Poaceae and was originated in Mexico and Central America. It possesses somatic chromosome number of 20, a genome size of 2.3 gigabase and more than 32,000 genes (Schnable et al. 2009). Maize grows well in various agroecologies and is unparalleled to any other crop due to its ability to adapt in diverse environments. It has emerged as a crop of global importance owing to its multiple end uses as a human food and livestock feed and serves as an important component for varied industrial products. Besides, maize serves as a model organism for biological research worldwide. Globally, about 1016.73 million metric tonnes of maize is produced every year – the highest among major staple cereals (FAOSTAT 2013). A major portion of maize produced worldwide is used for animal consumption as it serves as a vital source of proteins and calories to billions of people in developing countries, particularly in Africa, Mesoamerica and Asia (Shiferaw et al. 2011). Further, it is a source of important vitamins and minerals to the human body. Along with rice and wheat, maize provides at least 30 % of the food calories to more than 4.5 billion people in 94 developing countries. Maize provides over 20 % of total calories in human diets in 21 countries and over 30 % in 12 countries that are home to a total of more than 310 million people (Shiferaw et al. 2011). At present, the developed world uses more maize than the developing world, but forecasts indicate that by the year 2050, the demand for maize in the developing countries will double owing to the rapid growth in poultry industry, the biggest driver of growth in maize production (Rosegrant et al. 2009; Prasanna 2014). Improved maize

hybrids with substantial increase in production per unit area are required to feed the ever-growing population. Further, with changing climatic conditions, several new biotic stresses have emerged and minor disease and insect pests have become more prevalent and started inflicting more damages. Among abiotic stresses drought, heat and water logging are the major one and their simultaneous occurrence are now more frequent than ever. Malnutrition caused by deficiency of minerals and vitamins, especially iron, zinc and vitamin A, has been identified as one of the most important problems that require urgent attention worldwide (Bouis et al. 2011; Gupta et al. 2015a, b, c). Germplasm including wild relatives and landraces possess enormous potential as genetic resource for harbouring important and novel alleles/genes. These valuable germplasm can be systematically and effectively utilized in the crop improvement programmes worldwide, to develop high yielding and nutritious maize with resilience to biotic and abiotic stresses. Here we present origin and spread of maize, importance of wild relatives and landraces in the maize improvement programme.

4.2 Origin of Maize

The term ‘maize’ seems to be derived from the word ‘mahiz’ of Taino language of the Caribbean islands, which became ‘maiz’ in Spanish (Oxford dictionary 2015). Based on this common name, Linnaeus included the name as species in the botanical classification of *Zea*. Maize is also popularly known as ‘corn’ in English-speaking countries. In some countries, ‘corn’ means the ‘local staple’, while in some others it is used for any ‘cereal’. The ear of maize is unique among

cereals, and morphologically similar wild progenitor of maize could not be found. Therefore, its evolution has been a great scientific challenge and of great interest for both biologists and archaeologists. Many hypotheses/theories have been proposed by different scientists to explain the origin of maize. Among them, (1) tripartite hypothesis, (2) catastrophic sexual transmutation theory, (3) *Tripsacum-Zea diploperennis* hypothesis, and, (4) teosinte hypothesis were debated and discussed in detail by different scientists.

4.2.1 Tripartite Hypothesis

Tripartite hypothesis, proposed by Mangelsdorf and Reeves (1938, 1939), and later elaborated by Mangelsdorf (1974), states that maize was domesticated from some unknown wild, now extinct maize plant that had structures similar to the ear of modern maize. The hypothesis comprised three parts; (1) the progenitor of maize was a wild maize prototype from South America, which has become extinct or remained undiscovered; (2) teosinte is the offspring of a cross between maize and *Tripsacum*; and, (3) sections of *Tripsacum* chromosomes had contaminated maize germplasm. Thus, Mangelsdorf and Reeves explained the extreme morphological differences between maize and teosinte by imagining a missing ancestor, while relied on *Tripsacum* to explain their similarities. The evidence/validation for this explanation came from their own experiment. Mangelsdorf and Reeves obtained few sterile maize-*Tripsacum* hybrids from the artificial crossing. They could identify a minimum of four factors with large effect that controlled defined morphological syndrome. They believed that each of these four factors with large effects represented a block of many linked genes. This interpretation was tied to their hypothesis that maize evolved in nature from a now extinct form of wild maize over a long evolutionary period. They believed that the differences between maize and teosinte were very large, and hence it is not possible that teosinte was transformed into maize in the few 1000 years during which virtually all crops were domesticated. Until the 1960s, the tripartite hypothesis was

widely accepted. However, data were not sufficient and their hypothesis could not withstand the test of time (Doebley 2001).

4.2.2 Catastrophic Sexual Transmutation Theory

Ilitis (1983) proposed that maize was originated due to a sudden sexual transmutation that condensed the branches of teosinte and placed them in the female expression area of the plant. It states that the ear of maize was derived from the central spike of the tassel of teosinte. According to Ilitis, this has happened due to a phenomenon known as ‘genetic assimilation’. This resulted in substantial alterations in the nutrient distribution of the plant and led to drastic morphological changes. Morphogenetic and structural imbalance possibly had led to the transformation into primitive maize. During the late 1980s, teosinte hypothesis started gaining importance and the catastrophic sexual transmutation theory became less convincing.

4.2.3 *Tripsacum-Zea diploperennis* Hypothesis

Tripsacum-Z. diploperennis hypothesis can be considered as a modern version of the tripartite hypothesis and was given by Eubanks (1995). It proposed that maize arose from the progeny of a cross between *Z. diploperennis* and *Tripsacum dactyloides* (Eubanks 2001). This proposal was put forward with the observations on two putative hybrids viz. ‘Tripsacorn’ and ‘Sundance’ originated from these two grasses. The rudimentary ear of these putative hybrids had exposed kernels attached to a central rachis or cob. If such hybrids occurred naturally, then at least – according to proponents of the hypothesis – the evolutionary puzzle of the origin of maize and its unparalleled architecture is solved. Thus, this hypothesis challenged the idea that maize is a domesticated form of teosinte. However, *Tripsacum* and *Z. diploperennis* could not be hybridized successfully. The chromosome number of both ‘Tripsacorn’ and ‘Sundance’ is $2n=20$. These hybrids would be

expected to have 28 or 46 chromosomes if *Tripsacum* ($2n=36$ or 72) had indeed been one of the parents. Of the polymorphisms identified by RFLP data (Eubanks 1997), ‘Tripsacorn’ and ‘Sundance’ shared four times as many bands with *Z. diploperennis* as with *Tripsacum*, indicating a much closer relationship with teosinte than with *Tripsacum*. Besides, 23 % of the molecular markers surveyed were not found in either of the parents.

4.2.4 Teosinte Hypothesis

The teosinte hypothesis states that teosinte is the sole progenitor of maize. As proposed by Beadle (1939, 1972, 1978), the teosinte hypothesis states that (1) teosinte provided a useful food source and ancient people cultivated it; (2) during the cultivation of teosinte, mutations that improved teosinte’s usefulness to humans arose and were selected by people; (3) as few as five major mutations would be sufficient to convert teosinte into a primitive form of maize; (4) different mutations controlled different traits, viz. one mutation would have converted the disarticulating ear-type of teosinte into the solid ear type of maize; and, (5) over the period of time, humans selected additional major mutations coupled with many minor ones. Beadle (1939) believed that missing ancestor is not needed to explain the origin. He could obtain completely fertile hybrids between maize and teosinte. He interpreted that these two species were conspecific and that they had diverged only recently. Beadle actually used Mangelsdorf and Reeves’s own data against them and suggested that their four factors might correspond to four major genes. Each of these major genes controlled a single trait that transformed teosinte into maize. In modern form of teosinte hypothesis, *Z. mays* ssp. *parviglumis* (wild Mexican grass teosinte) has been pinpointed as the likely progenitor. Further, maize arose through large changes in *parviglumis* – through artificial selection for specific traits. Most maize geneticists and evolutionists have now accepted that maize is a domesticated derivative of *parviglumis* (Bennetzen et al. 2001). *Z. mays* ssp. *parviglumis*

grows as a wild plant alongside of the Balsas river and hence commonly known as Balsas teosinte.

4.3 Characteristics of Teosinte and Maize

Though it is accepted that teosinte is the progenitor of maize, extreme morphological differences persist between teosinte and maize (Doebley and Stec 1991) (Table 4.1). Some reported that the morphological differences between maize and

Table 4.1 Differences among teosinte and maize for various morphological characteristics

Teosinte	Maize
1. Teosinte plants are branched and produce many ears	1. Maize plants produce a single upright stem with one or few ears
2. Terminal position of primary branch bears a tassel	2. Primary branch is modified into ears
3. The leaves along the lateral branches are fully formed and composed of leaf blade and sheath	3. Leaves of the lateral branch are modified into husks which cover the ear
4. Secondary lateral branch is modified into ears	4. Secondary lateral branches are extremely rare
5. Ears are covered loosely by a single or few husks	5. Ears are covered tightly by many husks
6. Each ear possesses only two kernel rows	6. Each ear possesses about 8–22 kernel rows
7. Ear possesses about 10–12 kernels	7. Ear bear about 250–500 kernels
8. Each kernel is sealed tightly in a stony casing or fruitcase	8. Each kernel is naked and not covered by any fruitcase
9. During development, out of two spikelets one is aborted, hence each fruitcase holds a single-spikelet	9. Maize evolution involved the derepression of the second spikelet primordium, hence there are two mature spikelets
10. At maturity, fruitcase having the kernel shatter and become the dispersal units	10. At maturity, kernels do not shatter, and remain attached with ears
11. Majority of teosintes possess varying degree of seed dormancy	11. Seeds of maize do not possess dormancy

teosinte exhibit quantitative variation while others reported Mendelian pattern. Based on the conclusions from various studies, the genes responsible for transformation of teosinte into maize might be categorized into three overlapping classes: (1) major loci with simple inheritance, phenotypic effects of which are affected to lesser degree by environment; (2) QTL having complex inheritance are influenced by both genetic background and environment; and, (3) QTLs whose effects are dependent to a greater extent on epistatic interactions. This type of QTLs could exist in teosinte with little or no contribution to the phenotypic variation (Lauter and Doebley 2002). The available facts indicate that most of the morphological changes brought about during maize domestication could be accounted by QTL of measurable additive effects. Doebley and Stec (1993) and Doebley et al. (1994) mapped QTLs governing key contrasting traits using two F₂ populations generated between teosinte and maize. Key morphological traits differing between maize and teosinte have been located on chromosome 1 L (plant architecture), chromosome 2S (ear rank), chromosome 4S (cupulate fruitcase), chromosome 1 L, 1S and 3 L (pattern of spikelet: single vs. paired). Ear disarticulation showed complex inheritance as the observations were not consistent in two populations. However, a few loci having major effects, like *teosinte branched1* (*tb1*) and *teosinte glume architecture1* (*tga1*), may have played a larger role in transforming teosinte to modern maize.

4.3.1 Teosinte Branched1 (*tb1*)

The *teosinte branched1* (*tb1*) is a mutant of maize for plant architecture or apical dominance, and maize plant homozygous for mutant *tb1* shows lateral branches like teosinte but terminating with sterile tassels (Doebley et al. 1995, 1997; Doebley 2004). This gene has been reported as class II member of TCP family of transcriptional regulators which is hypothesized to negatively control cell cycle genes and results in repressed organ growth (Cubas et al. 1999; Doebley 2004). Expression analysis showed that maize allele of

tb1 is expressed twice the level of the teosinte allele and confers repression of branching in maize (Doebley 2004). The difference between maize and teosinte architecture was not due to change in coding sequence of *tb1* gene, instead due to variations in regulatory region (Doebley et al. 1997; Wang et al. 1999). Phenotypic fine-mapping and population genetic analysis indicated that the selection force operated between -58.7 and -65.6 kb of the *tb1*-ORF, and haplotype analysis between maize and teosinte revealed that there are four fixed differences [two SNPs and two insertions, i.e. *Hopscotch* retrotransposon and *Tourist* miniature inverted-repeat transposable element (MITE)] between the sequences in this region. These four differences were grouped into two regions, i.e. proximal (*Hopscotch* and one SNP) and distal (*Tourist* and the other SNPs), based on point of recombination between them. Of these, *Hopscotch* insertion is the major causative polymorphism between maize and teosinte (Studer et al. 2011). Thus, modification of *tb1* expression was an important target of selection during maize domestication. Recently, Vann et al. (2015) reported the presence of *Hopscotch* element in *tb1* among some populations of teosinte and concluded that the role of *Hopscotch* differs between maize and teosinte. Analyses of *tb1* expression levels in teosinte with and without the *Hopscotch* element could possibly provide more lights on regulation of *tb1*.

4.3.2 Teosinte Glume Architecture1 (*tga1*)

Another major difference between maize and teosinte is the presence of cupulate fruitcase for which *teosinte glume architecture1* (*tga1*) was identified as a major locus. The allele *tga1* has been found to have pleiotropic effect, and it affects several distinct aspects of fruitcase development. This suggested that *tga1* is a regulatory locus and it might get activated after the decision to become female is made. The *tga1* activates the programmes for invagination of the internode, internode elongation, three-dimensional growth

of the glume, silica deposition, and the pattern of lignifications (Dorweiler et al. 1993; Dorweiler and Doebley 1997). Genetic fine mapping was used to map *tga1* to a ~6 kb segment between the marker *bnlg252* and *bm22.7* and further fine-mapped to a 1042 bp segment between SNP (-1024) and SNP (+18). BLAST searches revealed that *tga1* has homology to squamosa-promoter binding protein (SBP) family of transcriptional regulators. Six fixed differences between maize and teosinte were identified by DNA sequence analysis, of which five are SNPs at 5' end of the coding sequence affecting *tga1* expression and the sixth difference is the substitution of lysine (K) in teosinte to asparagine (N) in maize. This substitution might have a role in altered protein stability, translation efficiency or protein function and is hypothesized to be the causative site of maize and teosinte polymorphism (Wang et al. 2005).

4.4 Diffusion of Maize

The centre of origin for maize is the Mesoamerican region now called Mexico and Central America (Watson and Dallwitz 1992). Most historians are of the view that maize was first domesticated in the Tehuacan Valley of Mexico (Roney and Hard 2009). Although maize pollen grains as old as 80,000 years have been identified from Mexico City, archaeological records suggest that domestication of maize is at least 6000 years old. However, a study by Matsuoka et al. (2002) has demonstrated that maize arose from a single domestication in Southern Mexico about 9000 years ago rather than the multiple independent domestication models. Though different studies have proved that maize was domesticated from *Z. mays* ssp. *parviglumis*, one major contradictory observation is that maize cultivars closely related to *parviglumis* are available in the Mexican highlands, whereas *parviglumis* is generally endemic to the mid- to lowlands. Domestication of maize in the highlands could be explained by the presence of *parviglumis* in the highlands at the time of domestication (Matsuoka et al. 2002; Piperno 2003) but unlikely because of cooler and drier

climate existed at the time of domestication (Metcalf 2006; Piperno et al. 2007, 2009). The other possibility could be that the early domesticate might have spread from lowlands to highlands and subsequent diffusion from highlands replaced lowland populations (Piperno 2003), which is also unlikely as it needs the highland maize to outcompete existing well-adapted populations in lowlands (Eagles and Lothrop 1994). Most acceptable hypothesis that explained this paradox is by the presence of a second subspecies of annual teosinte, *Z. mays* ssp. *mexicana* in the highland along with maize. Maize and *mexicana* are cross compatible (Ellstrand et al. 2007), and gene flow from *mexicana* to maize has also been reported (Wilkes 1972; Matsuoka et al. 2002). In comparison to maize, *mexicana* is more closely related to *parviglumis* (Matsuoka et al. 2002), so gene flow from *mexicana* can serve as indirect means of genetic similarity between highland maize populations and *parviglumis*.

As stated above, *parviglumis* is endemic to lowland tropical short-day condition, so adaptation of maize to longer days was prerequisite for its diffusion from tropical to temperate regions (Matsuoka et al. 2002; Piperno et al. 2009; van-Heerwaarden et al. 2011). One of the major genes reported to modulate photoperiod sensitivity is *ZmCCT* which encodes a CCT domain-containing protein. Adaptation under long day conditions was possible due to an insertion of a CACTA-like transposon into the *ZmCCT* promoter that can suppress the *ZmCCT* expression. Tropical maize plants with this transposable element insertion has been selected for by mankind and accumulated over time as maize adapted to range of long-day environments (Yanga et al. 2013). This insertion event helped maize to diffuse it from its centre of origin and adapt to different photoperiodic conditions around the world. Initial diffusion of maize across the America was a slow process which took over thousands of years, but following the discovery of the new world by Columbus, rapid diffusion took place around the world in the past 500 years (Dubreuil et al. 2006).

Mir et al. (2013) conducted the first genetic study at global level to find out the actual

diffusion path of maize from America to different parts of the world by genotyping 799 different landraces collected around the globe using SSR markers. Initially to identify population structure within centre of diversity, 258 American landraces itself were classified into seven distinct clusters, viz. 'Northern US flints', 'Mexican highlands', 'Tropical lowlands', 'Middle North American', 'Northern South American', 'Middle South American' and 'Andes', and used as prior information to interpret which of these clusters served as source germplasm to different non-American landraces. Northern US flints germplasm spread out through Europe from Northern France eastwards and also contributed through hybridization to Pyrenean-Galician landraces (Mir et al. 2013). The same was also evident in admixed-ancestry of Portuguese landraces, suggesting a hybrid origin. Diffusion of maize germplasm in the Middle East and Eastern Africa took place from the Middle North-American maize germplasm, but introduction to Eastern Africa was a distinct independent event rather than diffusion from the Middle East. The ancestry of North Eastern Chinese germplasm is traced back to Middle North American maize germplasm. Throughout Eastern Asia, mainly along the coasts, maize ancestry is traced back to Mexican highlands cluster, with an initial introduction into Indonesia. Thereafter, diffusion took place northward and towards Japan. Diffusion of tropical lowlands cluster contributed to the ancestry of Southern Spanish maize and also Moroccan landraces. Maize at Western Asia from Nepal to Afghanistan also showed tropical lowland ancestry but Mexican ancestry becomes predominant in Southeast Asia. Other clusters also acted as ancestral to maize germplasm for different parts of the globe. They are Northern South American cluster for Southern European landraces, Northern South American and Middle South American cluster for Western sub-Saharan African landraces, and Middle South American for few Eastern Asian landraces. But no clear diffusion path was observed for Andean germplasm that could be due to its geographical isolation and less productive nature beyond their ecological niches (Gouesnard et al. 2002).

4.5 Wild Relatives

The wild relatives of crop plants constitute an increasingly important genetic resource for improving agricultural production and maintaining sustainable agroecosystems (FAO 1998). A crop wild relative (CWR) can be defined as 'a wild plant taxon that has an indirect use derived from its relatively close genetic relationship to a crop' (Maxted et al. 2006). The grass genus *Zea* comprises of seven genera with varying chromosome number and are classified into two groups, viz. old world group and new world group (Table 4.2). *Coix*, *Chionachne*, *Sclerachne*, *Trilobachne* and *Polytoca* belong to the old world group and originated from the Southeast Asia. New world group consists of *Zea* and *Tripsacum* and are native to Mexico and Central America. *Z. mays* ssp. *mays* is the only species of *Zea* with economic importance, and the other species of *Zea* are generally referred to as teosintes.

4.5.1 Teosintes

Teosintes are wild grasses mainly originated in Mexico and Central America and represent potential genetic resources for maize improvement. All species of teosinte can be crossed with maize (Wilkes 1977; Wang et al. 2008, 2014). The group of four species, viz. *Z. diploperennis*, *Z. nicaraguensis*, *Z. luxurians* and *Z. perennis*, and three subspecies of *Z. mays*, viz. *Z. mays* spp. *huehuetenangensis*, *Z. mays* ssp. *mexicana* and *Z. mays* ssp. *parviglumis*, are collectively called teosintes (Table 4.2) (Fukunaga et al. 2005). Among these, *Z. mays* spp. *mexicana* is annual type teosinte that spread across the central highlands of Mexico.

4.5.2 *Tripsacum*

Tripsacum commonly called as 'gamma grass' belongs to the secondary gene pool of maize as defined by Harlan and de Wet (1971). This genus consists of nearly 16 perennial and warm species which originated in Mexico and Guatemala but

Table 4.2 Wild relatives of maize and their origin

S. No.	Genus	Species	Place of origin	2n ^a
1.	<i>Zea</i>	<i>mays</i> ssp. <i>mays</i>	Mexico and Central America	20
2.		<i>mays</i> ssp. <i>mexicana</i>	Mexico	20
3.		<i>mays</i> ssp. <i>parviglumis</i>	Southern and Eastern Mexico	20
4.		<i>mays</i> ssp. <i>huehuetenangensis</i>	Guatemala	20
5.		<i>diploperennis</i>	Jalisco and Mexico	20
6.		<i>luxurians</i>	South Eastern Guatemala	20
7.		<i>nicaraguensis</i>	Nicaragua	20
8.		<i>perennis</i>	Jalisco and Mexico	40
9.	<i>Tripsacum</i>	<i>australe</i>	South America	36
10.		<i>cundinamarce</i>	Colombia	36
11.		<i>floridanum</i>	South Florida and Cuba	36
12.		<i>latifolium</i>	North and Central America	36
13.		<i>laxum</i>	Central America	36?
14.		<i>bravum</i>	Mexico	36, 72
15.		<i>zopilotense</i>	Meso-America	36, 72
16.		<i>maizar</i>	Meso-America	36, 72
17.		<i>andersonii</i>	Central America	64
18.		<i>dactyloides</i>	Mexico and Guatemala	36, 72, 108
19.		<i>intermedium</i>	Guerrero, Chiapas, Guatemala, Honduras	72
20.		<i>manisuroides</i>	Tuxtla Gutierrez, Chiapas and Mexico	72
21.		<i>jalapense</i>	Oaxaca, Chiapas, Guatemala, El Salvador	72
22.		<i>lanceolatum</i>	South West USA	72
23.		<i>pilosum</i>	North and Central America	72
24.		<i>peruvianum</i>	South America	72, 90, 108
25.	<i>Coix</i>	<i>aquatica</i>	South East Asia	10
26.		<i>poilanei</i>		10
27.		<i>lacryma-jobi</i>		20
28.		<i>gigantea</i>		20, 40
29.		<i>ouwehandii</i>		—
30.		<i>puellarum</i>		—
31.		<i>gasteenii</i>		—
32.	<i>Chionachne</i>	<i>koenigii</i>	South East Asia	20, 40
33.		<i>semiteres</i>		20
34.		<i>hubberdiana</i>		—
35.		<i>biaurata</i>		—
36.	<i>Sclerachne</i>	<i>punctata</i>	South East Asia	20
37.	<i>Trilobachne</i>	<i>cookei</i>	South East Asia	20
38.	<i>Polytoca</i>	<i>macrophylla</i>	South East Asia	20, 40
39.		<i>massii</i>		—
40.		<i>digitata</i>		—
41.		<i>javanica</i>		—
42.		<i>wallichiona</i>		—
43.		<i>cyanthopoda</i>		—

^a2n represents somatic chromosome number

are also distributed in the USA and South America, and some of these are also detected in Asian regions (Table 4.2) (Berthaud et al. 1995; Chaudhary et al. 2014). It possesses wide variation in chromosome number ($2n=36, 64, 72, 90, 108$) among different species. After teosinte, *Tripsacum* is used widely as a source of valuable traits in maize due to its wider adaptation and resistance to heat, drought and waterlogging stresses.

4.5.3 Asiatic Genera

Asiatic genera, namely, *Coix*, *Sclerachne*, *Polytoca*, *Chionachne* and *Trilobachne*, collectively called as ‘Oriental Maydeae’, are prevalent across Southern parts of Asia, viz. India, Myanmar, China and Malaysia (Bor 1960; Chaganti 1965). Based on botanical features, the Asiatic genera are classified into two major groups: first, *Coix* wherein the caryopsis is enclosed in modified spathe, and second, *Trilobachne*, *Chionachne*, *Polytoca* and *Sclerachne* with caryopsis enclosed in endurated lower glume. The members of the second group differ from each other with respect to nature of lower glume of fruit, kind of grain base, visibility of hilum and rachis over the length and bearing of inflorescence on plant (Hernard 1931; Nirodi 1955; Bor 1960; Venkateswarlu and Chaganti 1973; Arora 1977; Sachan and Sarkar 1985). The details of species occurrence and chromosome complements of respective genera are listed in Table 4.2.

4.5.3.1 *Coix*

Coix plants are erect, tall and of floating types, while the inflorescence is a fascicled axillary and terminal spiciform raceme (Venkateswarlu and Chaganti 1973; Sachan and Sarkar 1985). Considering fruitcase morphology, *Coix* separates into cultivated ‘soft-shelled’ types and the wild ‘hard-shelled’ varieties. Apart from common features of *Coix* genus, the each species of *Coix* is botanically different from each other for leaf and fruit traits. Among various species of *Coix*, *C. lacryma-jobi* is one of the ‘soft-shelled’

types popularly known as “job’s tear” because of its grain shape resembling the drop of tear (Arora 1977). It contains highest protein content among cereals and seed extract which possesses therapeutic value (Ottoboni et al. 1990). At present, it spreads across the warmer parts of world and parts of India (Himachal Pradesh, Rajasthan, Uttar Pradesh, Madhya Pradesh, Andhra Pradesh, Tamil Nadu, Kerala, Karnataka, West Bengal, Orissa and Northeastern states). It is known by different names in local languages in different Indian states such as giral, kara, koa sangti, gurgur, kesai, kasi, golugu, gleun and many others (Venkateswarlu and Chaganti 1973). Taxonomical varieties of the ‘job’s tear’ (*C. lacryma-jobi*) includes *typica*, *stenocarpa*, *mayuen* and *monilifer* that differ for kernel shape, size, colour and hardness; of these, *mayuen* is of economic importance and other three are wild types (Watt 1904). The grains of *C. lacryma-jobi* var. *mayuen* is utilized by Garo, Karbi and Naga tribes as peanuts, brewing beers and making pounded flours even green leaves for silage purpose (Hore and Rath 2007). Based on repetitive sequence analyses, *Coix* has been found to be close to sorghum genome as well (Cai et al. 2014).

4.5.3.2 Other Asiatic Genera

Trilobachne genus has fruit with trilobite lower glume, larger central spike, linear hilum, 2–3 axial female spikelets and mainly staminate terminal inflorescence, while *Polytoca* genus possesses male spikes at terminal portion and female or mixed spikes at lateral portion with invisible hilum and hollowed out base of the grain (Sachan and Sarkar 1985). *Trilobachne cookei* is an annual herb reported to be uncultivated. The overlapping margin of lower glume which enclosed rachis segment is present in *Sclerachne*, but on the contrary, *Chionachne* contains non-overlapping margin with visible rachis. The profuse tillers and hardwood stems of *C. semiteres* and *C. koenigii* are used as fodder and making huts, respectively (Janaki-Ammal 1945). Although all these four genera include several species, most of which are not cultivated (Watson and Dallwitz 1992).

4.6 Wild Relatives and Cross Compatibility Barriers

Introgressive hybridization technique has been used extensively to transfer alien genes through method of repeated backcrossing of hybrid derived either from teosinte or *Tripsacum* with maize (Mano and Omori 2013; Wang et al. 2014). While intra- and interspecific hybridization aim at widening the gene pool of crop plants vis-a-vis crop improvement, a combination of spatiotemporal and physiological factors involving the genetic complexity in resulting hybrids prevents indiscriminate hybridization. There are barriers to hybridizing different species or genotypes in maize, and the underlying phenomena are ‘genetic incompatibility’ and ‘genomic incongruity’ (Allard 1960). When normal congruous relationship between pollen and pistil is disrupted, fertilization fails to occur as pre- and postfertilization processes are not consummated due to incongruity (Hogenboom 1973). The incompatibility barrier to hybridization may be complete if no seed is set despite the use of viable pollen on potentially receptive silks. Partial compatibility is detected by differential pollen tube growth and the distorted recovery of alleles coupled with incompatibility genes. Both complete and partial incompatibilities have been reported in maize when crossed with wild relatives.

4.6.1 Cross Compatibility

In *Z. mays*, intraspecific hybridization is fully cross-fertile (Purseglove 1972), though there is great cross compatibility between maize and other *Zea* species such as annual teosinte, except for the tetraploid, *Z. perennis*, that exhibits low fitness (Wilkes 1977; Galinat 1988). Though they are cross compatible, the rate of spontaneous or natural hybridization of teosintes with maize is species-dependent. For instance, *Z. mays* ssp. *luxurians* rarely hybridizes, *Z. mays* ssp. *mexicana* naturally hybridizes at a low rate (1%), whereas *Z. mays* ssp. *parviglumis* hybridizes at a high rate (50%) with maize (Ellstrand et al. 2007). Intergeneric hybridization between *Zea*

species and *Tripsacum* species (*T. dactyloides*, *T. floridanum*, *T. lanceolatum*, and *T. pilosum*) is difficult, but hybrids can be realized successfully through manual crossing. The pioneer publications by Mangelsdorf and Reeves (1931) demonstrated the possibility of cross between maize and *Tripsacum*. Moreover, several researchers supported the possibility of gene transfer from *Tripsacum* to maize (Galinat 1973; Bernard and Jewell 1985). However, *Tripsacum* is not easily crossable to maize; hence, few pathways have been developed to accomplish successful gene transfer from *Tripsacum* to maize background (Savidan and Berthaud 1994; Leblanc et al. 1996). Since maize-*Tripsacum* hybrid plants are sterile in nature, ‘tripsacorn’ from a cross of *Tripsacum* and *Z. diploperennis* has been used as bridge species to accomplish gene transfer from *Tripsacum* to maize (Kindiger and Backett 1992; Eubanks 2006). However, these hybrids exhibit high pollen sterility with limited female fertility due to the unequal number of chromosomes across genera and lack of pairing between them (Eubanks 1997). *Tripsacum* has been primarily utilized in several studies to understand its role in maize evolution and to characterize the resistance genes. *T. dactyloides* has been the focus of extensive breeding work to transfer the apomictic reproductive nature from *Tripsacum* to maize (Kindinger et al. 1996). There are no reports on compatibility between maize and the genera belonging to the old world group such as *Coix*, *Sclerachne*, *Polytoxa*, *Chionachne*, and *Trilobachne*, and genetic studies do reveal that these genera are distinct from both maize and teosintes of the new world group (Katiyar and Sachan 1992).

4.6.2 Genetic and Molecular Basis

Understanding the molecular mechanisms of crossing compatibility barriers in wide hybridization is of great importance. These barriers may be classified into two types, viz. (1) pre-fertilization or pre-syngamous and (2) post-fertilization or post-syngamous barriers (Stebbins 1958; Gutierrez-Marcos et al. 2003). As they contribute

to species isolation and speciation, pre-fertilization barriers relating to pollen tube germination, growth, guidance and sperm delivery are considered the major hybridization controls in nature (Dresselhaus et al. 2011). However, in incompatible crosses, stigmatic surface proteins are not involved in incompatibility responses, since germination occurs normally before pollen tube reaches, and are arrested at or near the surface of the stigma (Heslop-Harrison and Heslop-Harrison 1982). So, the initial steps in the process of hybridization such as stigmata pollen capture, adhesion, hydration, and germination do not seem to represent a pre-fertilization barrier in maize and other related grasses (Dresselhaus et al. 2011). The first pre-fertilization barrier, which involves the invasion of the stigma by pollen tubes, begins with the crossing of the cuticle of the stigma and the invasion of the intercellular space of the sporophytic tissues and pollen tube guidance towards the transmitting tract (Heslop-Harrison 1982). The second pre-fertilization barrier appears to be pollen tube growth support inside the transmitting tract in maize. Intensive cell-cell interactions between the pollen tubes and the female tissues of the pistil take place for stimulating and guiding the pollen tube growth (Dresselhaus and Franklin-Tong 2013). The growth of *Tripsacum* pollen tubes in maize pistil is arrested after about 2 cm growth, probably after pollen resources are consumed (Lausser et al. 2010).

Three genetic loci responsible for governing the compatibility barriers have been identified in North American flint and dent maize varieties. The first genetic system, *Gal/Gal*, is partially dominant. Genotype with *gal/gal* generates pollen that cannot fertilize with *Gal-s/Gal-s*, but is partially receptive on *Gal-s/gal-s* silks (Nelson 1952) and fully fertile in reciprocal crosses (House and Nelson 1958). Further, the two other genetic systems that have been described are the *Teosinte crossing barriers1* (*Tcb1*) and the *gametophytic factor2* (*ga2*) loci that restrict the crossability of maize pollen to silks of some annual maize subspecies teosinte (*Z. mays* ssp. *mexicana*) populations (Evans and Kermicle 2001; Kermicle 2006; Kermicle and Evans 2010). In

vivo assays of the dynamics of incompatible pollen tube growth and morphology in silks carrying the *Tcb1-s*, *Gal1-s* or *Ga2-s* barriers showed slower pollen tube growth than in compatible crosses at early stages and growth ceased 24 h after pollination (Lu et al. 2014).

Cross-compatible barriers in most intergeneric crosses prevent the formation of hybrid zygotes, as the pollen tubes are usually arrested in the proximity of or inside the transmitting tract (Heslop-Harrison 1982; Lausser et al. 2010). There are reports that chemotactic factors or molecules are secreted by the transmitting tract cells, to guide self, but not alien pollen tubes in maize and other grass species. A large percentage of *Tripsacum* pollen tubes are mistargeted to the stigma cortex instead of reaching the transmitting tract of maize silks, whereas in the reciprocal cross maize pollen tubes occur exclusively in the transmitting tract of *Tripsacum* (Lausser and Dresselhaus 2010; Lausser et al. 2010). In the case of a cross between maize and *C. lacrym-jobi* var. *frumentacea*, the pollen tubes could not reach the base of the maize style leading to strict reproductive isolation. The ratio of abnormal pollen tubes in the cross between maize and *Coix* was higher than that between maize and *Tripsacum*, suggesting that maize is closer to *Tripsacum* than to *Coix* (Duan et al. 2008).

Post-fertilization barriers arise due to hybrid embryo abortion or failure in endosperm development. In the cross between maize and *Tripsacum*, pollens of *Tripsacum* could germinate and release androgamete into embryo sac of maize, but normal hybrids were not obtained because of embryo sac incompatibility or embryo abortion (Duan et al. 2008). Endosperm development is regulated by specific balance of gene expression and epigenetic imprinting mechanisms (characterized by their uniparental expression) from either the maternal or paternal genomes. Any alteration in this balance resulting from changes in allelic copy number, sequence or epigenetic imprints can cause endosperm failure and eventual embryo abortion (Gutierrez-Marcos et al. 2003). Histological studies on hybrids of maize $\times$ *Tripsacum* showed abnormalities of endosperm development at very early stages

ranging from defects at the chalazal pole (Cooper and Brink 1942) to abnormal free nuclear division rates (Brink and Cooper 1940).

Hybrid sterility, abnormal male and female gametes in hybrid, can also be considered as a barrier in wide hybridization. Disturbances in pollen development of maize $\times$ *T. dactyloides* hybrids, derived from both diploid and tetraploid *Tripsacum* sources, are commonly observed due to anomalies in the development of the microspore. The anomalies observed are failure of the chromosomes to congregate at the metaphase plate, development of a tripolar spindle and failure of cytokinesis at the first and second meiotic divisions (Kindiger 1993). Pre-hybridization barriers are typically stronger than post-hybridization barriers in establishing discriminate hybridization in plants (Rieseberg and Willis 2007). Even though the mechanisms on how a crossing barrier arises in sexually reproducing plants are largely unknown (Lu et al. 2014), the possibilities of wide crosses and successfully introgressing desired traits from wild relatives into maize are nevertheless limited; however, different approaches can be employed to overcome these barriers.

4.7 Wild Relatives in Maize Improvement

With the advent of climate change and greater ecosystem instability, CWRs are likely to prove a critical resource in ensuring food security in the coming years. The wild maize germplasm contains significant variation for plenty of economically important traits that have remained untapped (Hoisington et al. 1999). The alien gene introgressions especially from teosinte and *Tripsacum* possess enormous potential for maize improvement to (1) increase agronomic performance, (2) transfer genes for quality traits, (3) introgress disease/pest resistance genes and (4) improve level of tolerance to abiotic stresses. Some of the traits that have been improved through wide hybridization of maize with teosinte and *Tripsacum* are presented (Table 4.3).

4.7.1 Improvement of Agronomic Performance

Cohen and Gallinat (1984) have described the importance of wild chromosome segments of teosinte and *Tripsacum* for improving maize agronomic performance. The upscaling of quantitative traits in cultivated maize has been achieved by using sexual hybridization to teosinte and *Tripsacum* (Ramirez 1997). Several experiments showed the better performance of introgressed maize lines produced from utilizing teosinte for yield, tillering and green matter production, over the maize hybrid checks (Pasztor and Borsos 1990; Savidan and Berthaud 1994). SD00100, an introgression line selected from the maize $\times$ mexicana population, possesses big ears and stay-green characters (Wang et al. 2008). Petrov (1984) suggested the development of apomictic maize genotypes by introgressing apomixis genes from *T. dactyloides* to maize. Consequently, many studies were conducted for this purpose (Table 4.3). Furthermore, bulked segregant analysis performed by Leblanc et al. (1995) identified molecular markers linked to diplospory in *T. dactyloides* and showed that RFLP marker on long arm of chromosome 6 cosegregates with diplosporous mode of apomixis. Later, efforts were made to transfer apomixis from diplosporous tetraploid *Tripsacum* into maize through conventional backcrossing (Kindiger et al. 1996; Leblanc et al. 1996).

4.7.2 Enhancement of Quality Attributes

Wild relatives have been utilized for enhancement of quality traits in maize. Introgressed maize genotype from maize $\times$ teosinte possesses better silage quality over hybrid checks (Sidorov and Shulakov 1962). Introgression of *Z. mays* ssp. *mexicana* into maize resulted in lines with higher protein content, as well as higher in amino acids such as lysine, methionine and/or phenylalanine (Table 4.3) (Pasztor and Borsos 1990; Wang et al. 2008). Angiotensin converting

Table 4.3 List of some of the economically important characters introgressed from teosinates and *Tripsacum*

S. No.	Characters	Donor genera/species	Reference(s)
Agronomic traits			
1.	Higher yield, profuse tillering, biomass production, increased height	<i>Z. mays</i> spp. <i>mexicana</i>	Pasztor and Borsos (1990) and Wang et al. (2008)
2.	Apomixis	<i>Tripsacum dactyloides</i>	Petrov (1984), Savidan and Berthaud (1994), Leblanc et al. (1995, 1996), and Kindiger et al. (1996)
Quality traits			
4.	Superior silage making ability	Teosinte	Sidorov and Shulakov (1962)
5.	Higher protein, amino acid balance	<i>Z. mays</i> spp. <i>mexicana</i>	Pasztor and Borsos (1990) and Wang et al. 2008
6.	Reduced hypertension	<i>Z. mays</i> spp. <i>mexicana</i>	Wang et al. (2014)
Biotic stress resistance: <i>Disease-pests</i>			
7.	Anthracnose, Northern corn leaf blight, Southern corn leaf blight, Stewart's bacterial wilt, Common rust	<i>T. dactyloides</i>	Bergquist (1979, 1981)
8.	Northern corn leaf blight	<i>T. floridanum</i>	Hooker and Perkins (1980)
9.	Northern corn leaf blight, Southern corn leaf blight	<i>Z. diploperennis</i>	Wei et al. (2003)
10.	Northern corn leaf blight	Teosinte	Ott (2008)
11.	Corn smut	Teosinte	Chavan and Smith (2014)
12.	Downy mildew	<i>Z. mays</i> spp. <i>mexicana</i> , <i>Z. diploperennis</i>	Ramirez (1997)
13.	Ear rot	<i>Z. mays</i> spp. <i>mexicana</i>	Pasztor and Borsos (1990)
	Maize chlorotic dwarf virus	<i>Z. diploperennis</i>	Findley et al. (1983)
Biotic stress resistance: <i>insect pests</i>			
14.	Corn borer	<i>Z. mays</i> spp. <i>mexicana</i>	Pasztor and Borsos (1990)
15.	Asiatic corn borer	<i>Z. mays</i> spp. <i>mexicana</i> , <i>Z. perennis</i> , <i>Z. diploperennis</i>	Ramirez (1997)
16.	Corn rootworm	<i>T. dactyloides</i>	Eubanks (2002, 2006), and Prischmann et al. (2009)
Biotic stress resistance: <i>parasitic weeds</i>			
17.	Striga	<i>Z. diploperennis</i>	Menkir (2006), Amusan et al. (2008), and Yallou et al. (2009)
Abiotic stress tolerance			
18.	Flooding	<i>Z. luxurians</i> , <i>Z. nicaraguensis</i> , <i>Z. mays</i> ssp. <i>huehuetenangensis</i>	Ray et al. (1999), Mano et al. (2005), and Mano and Omori (2007, 2013)
19.	Salinity	<i>T. dactyloides</i>	Pesqueira et al. (2003, 2006) and Shavrukov and Sokolov (2015)

enzyme (ACE) is one of the key regulators of blood pressure, and ACE inhibitors (ACEI) are widely used to treat hypertension in human. Drugs such as captopril and enalapril possess various side effects and naturally found ACEI can be potentially utilized as the drug (Vyssoulis et al. 2001). Introgression line SD00100 generated from a cross between maize and *Z. mays* ssp. *mexicana* possesses high ACEI peptides (Wang et al. 2014). This novel inbred line thus can be utilized as functional foods or nutraceuticals in the near future.

4.7.3 Resistance to Biotic Stresses

Scientists have introgressed genes imparting resistance to various diseases, viz. anthracnose, northern corn leaf blight, southern corn leaf blight, bacterial wilt, common rusts, common smut, downy mildew, ear rot and maize chlorotic dwarf virus, from teosinte and *Tripsacum* to cultivated maize through typical sexual hybridization (Table 4.3) (Bergquist 1979, 1981; Hooker and Perkins 1980; Findley et al. 1983; Pasztor and Borsos 1990; Ramirez 1997; Wei et al. 2003; Ott 2008; Chavan and Smith 2014) (Table 4.3). Further, resistance against corn borers and root worms have been introgressed into maize from wild relatives (Pasztor and Borsos 1990; Ramirez 1997; Eubanks 2002, 2006; Prischmann et al. 2009). Striga is an obligate root hemiparasite of cereals including maize spread in regions of Africa and parts of Asia. Striga development is dependent on exudates and solutes of host roots that affect host plant health (Rich and Ejeta 2008). For the development of host plant resistance, Lane et al. (1997) reported *Z. diploperennis* as the source of striga-resistant genes for maize improvement, while Gutierrez-Marcos et al. (2003) reported *T. dactyloides* as secondary source for striga resistance in maize. Rich and Ejeta (2008) highlighted the importance of both sources over the landraces for the development of striga-resistant maize in African regions. The scientists of International Institute of Tropical Agriculture (IITA) have developed ZD05, *Striga hermonthica*-resistant inbred from *Z. diploperennis* accession and

tropical maize germplasm (Menkir 2006; Amusan et al. 2008; Yallou et al. 2009).

4.7.4 Tolerance to Abiotic Stresses

Maize succumbs to yield loss by abiotic factors such as drought, heat, flooding and salt due to limited availability of cultivated genotypes possessing tolerance to these abiotic factors (Table 4.3) (Wilkes 1967; Ramirez 1997; Chaudhary et al. 2014). Therefore, search for genes for abiotic stress tolerance has been diverted towards wild germplasm. The teosintes especially *Z. nicaraguensis*, *Z. luxurians* and *Z. mays* ssp. *huehuetenangensis* are well-characterized for its high tolerance for flooding on account of more adventitious roots above the soil surface and well-formed aerenchyma in adult plants (Ray et al. 1999; Mano et al. 2005; Mano and Omori 2007, 2013). Similarly, *T. dactyloides* was reported as tolerant species to excess salt conditions through screening of maize × *Tripsacum* hybrids for survival on exposure to NaCl concentrations (Pesqueira et al. 2003, 2006). Likewise, studies have confirmed the chromosome segment belonging to *Z. mays* ssp. *huehuetenangensis*, *Z. nicaraguensis* and *Z. luxurians* have led to flooding tolerance in maize (Mano and Omori 2007, 2013). Recently, Shavrukov and Sokolov (2015) developed lines with salinity tolerance from maize × *T. dactyloides*.

4.8 Landraces of Maize and Its Use in Crop Improvement

As defined by Camacho-Villa et al. (2005), a landrace is ‘dynamic population(s) of a cultivated plant that has historical origin, distinct identity, and lacks formal crop improvement, as well as often being genetically diverse, locally adapted, and associated with traditional farming systems’. Maize possesses tremendous genetic diversity compared to other crops that offers potential opportunities for the improvement of maize for its grain yield, tolerance to biotic and abiotic stresses and specialty traits. In landraces, variation tends

to be partitioned within populations, rather than between populations, and levels of variation can be very high. Landraces worldwide have been characterized using various morphological characters. With the advent of DNA-based markers, characterization using molecular markers is now the preferred choice, mainly due to their abundance and environment-neutral behaviour (Prasanna 2010, 2012). However, landraces being heterogeneous in nature, large number of individuals in each of the accessions are required to be genotyped and the use of molecular markers are very expensive and time-consuming (Prasanna 2012). A new method for SSR analysis has been formulated in which genotyping of pools of individuals from a population than genotyping of multiple individuals per population is very effective and accurate (Dubreuil et al. 2006). A minimum of 15 individuals are required to be characterized to adequately represent the allelic diversity present in a population (Dubreuil et al. 2006; Warburton et al. 2010). This bulk DNA fingerprinting technique using SSR markers developed at CIMMYT has been used to differentiate the improved open-pollinated varieties (OPVs) or synthetics (Warburton et al. 2010). Large number of studies have undertaken molecular characterization of landraces to understand the extent of genetic diversity and their relationships (Barcaccia et al. 2003; Carvalho et al. 2004; Qi-Lun et al. 2008; Sharma et al. 2010; Singode and Prasanna 2010; Warburton et al. 2011; Wasala and Prasanna 2013; Molin et al. 2013; Semagn et al. 2014; Oppong et al. 2014). The natural variability of landraces has been exploited for various biotic stress resistance, abiotic stress tolerance and quality improvement of maize (Prasanna 2010, 2012), apart from understanding the diffusion routes of maize. Some of the examples of their utilization are as follows.

4.8.1 Resistance to Biotic Stresses

Because of vast geographical distribution, maize encounters a large number of diverse pest and pathogens during its life-span. The foliar diseases like turicum leaf blight (TLB) and maydis leaf

blight (MLB) are severe constraint for maize productivity throughout the world. Tuxpeno Crema a subpopulation of landrace Tuxpeno from Mexico is well known for its resistance to tropical foliar diseases (Rodriguez et al. 1998). A popcorn landrace, Palomero Toluqueno, was found to have resistance to the maize weevil, *Sitophilus zeamais* (Arnason et al. 1994). Few Caribbean landraces were found tolerant to larger grain borer (*Prostephanus truncatus*) (Kumar 2002).

4.8.2 Tolerance to Abiotic Stresses

Some of the Mexican maize landraces with abiotic stress tolerance comprise La Posta Sequia, Conica, Conica Nortena, Bolita, Breve de Padilla, Nal Tel, Tuxpeno (drought tolerant), Oloton (acid soil tolerant) and Chalqueno × Ancho de Tehuacan cross (alkalinity tolerant) (Prasanna 2012). Drought-tolerant population-1 (DTP-1) and drought-tolerant population-2 (DTP-2) developed for drought tolerance at CIMMYT are being exploited for development of inbred lines. DTP-1 is developed by 74 half-sib crosses of 13 drought-tolerant sources, while DTP-2 was constituted by introgression of 25 new drought-tolerant landraces in DTP-1 by half-sib crossing and further recombination (Edmeades et al. 1996). Some of the derived lines from La Posta Sequia were also found to be tolerant to both drought and heat stresses (Cairns et al. 2013).

4.8.3 Improvement of Specialty Traits

There are several landraces, viz. Pepitilla, Bolita, Azul, Tlacoya and Oaxaqueno, which are very popular and preferred for their tortilla quality. One of the prolific landrace, Murlimakai, from the Northeastern states of India has been exploited for development of a baby corn composite 'VL Baby Corn' (Prasanna 2012). The sequencing of Palomero, a popcorn landrace in Mexico, would further boost the understanding of popping quality improvement as well as structure of landraces (Vielle-Calzada et al. 2009).

4.8.4 Source for Genetic Diversification

In 1987, Latin American Maize Programme (LAMP) was initiated for generating information by evaluating landraces in Latin American countries with an objective to evaluate over 14,000 accessions found in Latin America and the US germplasm banks (Salhuana et al. 1991). Along with yield evaluation, landraces were identified for lodging tolerance, earliness and plant characteristics. The subset of landraces was made available to breeders to broaden the genetic base for hybrid development. Later, it was realized that LAMP materials had to be enhanced for utilization in breeding programme, as those germplasm were not adapted to temperate corn belt of America. During the time, the US-germplasm enhancement of maize (US-GEM) was conceived with the idea of introgression of valuable genes. This project, a collaboration among public and private institutions, aims at enhancing productivity and improving diversity of maize in the USA. It was initiated with improvement of 24 tropical and 27 temperate accessions identified on the basis of yield performance in LAMP; furthermore, seven hybrids donated by DeKalb Genetics have been included (Salhuana and Pollak 2006). The recent initiative of CIMMYT, ‘Seeds of Discovery’ (SeeD), was launched with an aim to find the allelic variation in the genetic resources of maize and wheat. This programme utilizes high-density genotyping, phenotyping and bioinformatic tools to know the desirable alleles and haplotypes linked with priority traits to be used by breeders (Prasanna 2012).

4.8.5 Understanding Diffusion Routes

Landraces are grown by the farming community at specific places for a long period of time. Unique diversity preserved in specific landraces help in tracking the insights into domestication events in maize (Matsuoka et al. 2002) and the migration routes of maize from the centres of origin (Rebourg et al. 2003; Vigouroux et al. 2008).

258 landrace accessions from America, 237 from Africa, 13 from Middle East, 148 from Europe and 143 from Asia were characterized using SSR markers for understanding the diffusion routes of maize worldwide (Mir et al. 2013). This was carried out as a part of the Generation Challenge Programme (GCP), involving researchers from CIMMYT-Mexico, INRA-France, IITA and national programmes of China, India, Indonesia, Thailand, Vietnam and Kenya. This led to the assessment of genetic relationships among landraces/populations worldwide and understanding of global diffusion of maize from primary centre of origin (Mir et al. 2013). Besides, landraces provided a clear understanding of phylogenetic relationships and gene flow between maize landraces and the wild progenitor, teosinte (Warburton et al. 2011; van-Heerwaarden et al. 2011).

Sequencing of the most popular and widely used inbred, B73 (Schnable et al. 2009) and Palomero, a popcorn landrace in Mexico (Vielles-Calzada et al. 2009), are important landmarks in maize genome research. With the advances in high-throughput SNP-based genotyping facilities and reduced cost of genotyping, it would be possible to understand the evolution of maize at finer level. The Palomero genome is smaller (22%) than that of B73 genome and shows a large number of unreported sequences, indicating the presence of a large number of unexplored alleles. More than 12 genes related to heavy metal detoxification and environmental stress tolerance were found to be conserved in B73 and Palomero but are absent from teosinte. This suggests that these genes were possibly involved in the domestication process during the evolution of maize (Vielles-Calzada et al. 2009).

Further, ‘Sikkim Primitive’, a landrace of high altitude regions of Northeastern Himalayan State of India, Sikkim, has been the centre of attraction among the maize breeders for decades (Sachan and Sarkar 1986). Locally known as ‘Murli makai’, the landrace is characterized by the absence of apical dominance, high prolificacy (having five to nine uniform ears per plant), erect leaves, tall with drooping tassel, popcorn type kernels and top bearing habit (Sachan and Sarkar 1982). It also has stay-green character, which

make it suitable for fodder use. Due to its primitive features, the landrace was phrased as ‘Sikkim Primitive’ (SP1 and SP2) (Dhawan 1964). Though ‘Sikkim Primitive’ has been characterized using cytological, biochemical and molecular markers (Sachan and Sarkar 1982, 1986; Bhat et al. 1998; Sharma et al. 2010), its utilization in the maize breeding programme has been greatly limited due to its poor expression outside the Northeastern Himalayan State (Sharma et al. 2010). Sequencing of ‘Sikkim Primitive’ landrace would further provide insight into the genome architecture, the regulation of key genes responsible for its primitive nature, and would in turn help in understanding its evolution and devising strategies for better utilization of the valuable landraces in the breeding programme.

4.9 Conclusions

Wild relatives and landraces of maize possess enormous potential as donors for important alleles/genes. However, these unexplored otherwise valuable germplasm have been utilized to a less significant extent in the breeding programmes. The inbreds derived from these genetic resources may not be directly utilized in the breeding programme; however, they can serve as important donors for specific traits. Transfer of genes from wild relatives is not always straight forward, and utilization of bridging species and special techniques such as embryo rescue may provide solutions. Comprehensive evaluation of landrace accessions for biotic/abiotic stresses and nutritional quality attributes would provide opportunity to the breeders to utilize them in the specific breeding programme. Effective use of available genome sequences of ‘B73’ and ‘Palomero’ landrace, coupled with strong ‘pre-breeding programme’, can potentially enhance the utilization of these unexplored germplasm in the maize breeding programme.

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